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Synopsis

On

ASCA-Fair Donation Allocation

Research Project Synopsis submitted to

Masters of Computer Application (SEM-III)

Of

University of Mumbai

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BATCH: 2025- 2027

Content of the Synopsis

1. Title - ASCA – An Integrated NGO and Donation Management System

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2. Introduction

ASCA is a web-based NGO and donation management system designed to digitally connect donors, NGOs, administrators and delivery personnel through a centralized platform. The system aims to simplify the process of registering users, managing donations, coordinating NGO activities and tracking the movement of donations.

The platform provides separate role-based access so that each stakeholder can perform activities according to their responsibilities. Donors can view and manage their donations, NGOs can manage donation requirements and received donations, delivery personnel can handle assigned deliveries, while administrators can monitor and manage the overall system.

The system is designed to improve transparency, coordination and efficiency in donation management while reducing dependency on manual records and communication.

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3. Problem Statement

Traditional NGO and donation-management activities often depend on manual records, separate communication channels and inefficient coordination between donors, NGOs and delivery personnel. This can make it difficult to track donations, manage requests, monitor delivery status and maintain transparent records.

Therefore, there is a need for a centralized web-based system that provides role-based access, structured donation management, delivery coordination and centralized monitoring for different stakeholders involved in the donation process.

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4. Objectives

The main objectives of ASCA are:

- To develop a centralized web-based platform for NGO and donation management.
- To provide separate role-based access for Admin, NGO, Donor and Delivery Personnel.
- To facilitate digital management of donation requests and donation records.
- To improve coordination between donors, NGOs and delivery personnel.
- To provide tracking of donation and delivery status through a structured workflow.
- To reduce dependency on manual record keeping and communication.
- To improve transparency and accessibility of donation-related information.

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5. Existing System / Existing Work

Existing digital donation systems provide facilities such as online donations, donor management, NGO management and donation tracking. Recent research has also explored blockchain-based systems to improve transparency and traceability in donation transactions. However, many systems concentrate primarily on financial donations, blockchain-based transparency, or a specific donation domain.

Recent IEEE work such as KindNest (2025) proposes a platform connecting donors, NGOs and volunteers, with features including donation listing, request management, volunteer scheduling, communication and real-time donation tracking.

Similarly, research on real-time donation management has highlighted problems such as data redundancy, delayed responses, inefficient matching and lack of real-time updates, demonstrating the need for centralized and coordinated digital systems.

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6. Literature Review

Paper 1 – KindNest – An Essentials Connect, 2025

The proposed system connects donors with people in need through NGOs and provides features such as donation listing, request management, volunteer scheduling, communication and donation tracking. The study focuses on improving transparency and coordination between donors, NGOs and volunteers.

Paper 2 – Real-Time Blood Donor Prediction and Campaign Optimization using DNN-Autoencoder and MERN Architecture, 2025

This research identifies problems such as data redundancy, inefficient donor matching, delayed responses and lack of real-time updates. It proposes a web-based system with authentication, dashboards and notifications to improve coordination and management.

Paper 3 – A Practical Decentralized Donation Based Crowdfunding Platform Using Blockchain, 2025

This study investigates blockchain-based donation platforms for improving transparency, security and efficiency. It highlights the problem of trust and transparency in centralized donation platforms and proposes decentralized donation management using blockchain.

Paper 4 – Blood Delivery Management System using Google Map – Blood Hub, 2024

The study proposes a web/mobile platform for connecting donors and recipients in real time. It identifies limitations in existing systems including lack of real-time data, incomplete donor information and coordination difficulties.

Literature Review Conclusion

The reviewed studies demonstrate that digital platforms can improve transparency, real-time coordination, tracking and management in donation-related activities. However, there remains scope for a simpler centralized platform that integrates different stakeholders and their responsibilities into a single role-based system.

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7. Gaps Identified / Research Gap

Based on the reviewed literature, the following gaps were identified:

- Many existing systems focus on a specific type of donation rather than general donation management.
- Some systems primarily focus on blockchain-based transparency, making them comparatively complex for a basic NGO implementation.
- Donor, NGO and delivery activities are not always integrated into a single centralized workflow.
- Manual coordination can cause delays in donation processing and delivery.
- There is a need for clear role-based access for different stakeholders.
- Donation and delivery status can be difficult to monitor when information is maintained through separate systems or communication channels.

Proposed Solution

ASCA addresses these gaps through a centralized role-based platform integrating Admin, NGO, Donor and Delivery Personnel into one system. It provides structured donation management, request handling and delivery coordination without introducing unnecessary technical complexity.

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8. Proposed Methodology / System Overview

The proposed system follows a centralized web-based architecture with role-based access.

Workflow:

Landing Page → User Login → Role-Based Dashboard → Donation/Request Management → Delivery Assignment → Status Update → Admin Monitoring

The system will provide separate functionality according to the user's role:

Donor

- Register/Login
- View donation requirements
- Submit/manage donations
- View donation status/history

NGO

- Manage NGO profile
- Add/view donation requirements
- Manage received donations

- Coordinate donation requests and deliveries

Delivery Personnel

- View assigned deliveries
- Manage delivery status
- Update delivery progress

Admin

- Manage users and roles
- Monitor NGOs and donors
- Monitor donations and deliveries
- Manage overall system activities

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9. Software Requirements

Component	Requirement
Operating System	Windows 10/11
Frontend	React.js / HTML / CSS / JavaScript
Backend	Python
Backend Framework	FastAPI
Database	SQL-based database
Development Environment	Visual Studio Code / AI-assisted development tools
Browser	Google Chrome / Microsoft Edge
Version Control	Git / GitHub
API Communication	REST API

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10. Hardware Requirements

Minimum Hardware

- Processor: Intel Core i3 or equivalent
- RAM: 4 GB minimum
- Storage: 10 GB available space

- Internet Connection: Required
- Display: Standard HD display

Recommended

- Processor: Intel Core i5 or equivalent
- RAM: 8 GB or higher
- Storage: 20 GB or more available space

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11. Scope

The scope of ASCA includes digital management of NGO-related donation activities through a centralized platform.

The system can be extended in the future with:

- Online payment gateway integration
- Real-time GPS-based delivery tracking
- Notifications through email/SMS
- Donation analytics and reports
- NGO verification/document management
- Mobile application
- AI-based donor/NGO recommendations

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12. Expected Outcome

The expected outcome of ASCA is a functional web-based platform that:

- Provides secure role-based access.
- Centralizes NGO and donation-related information.
- Simplifies donation and request management.
- Improves coordination between donors, NGOs and delivery personnel.
- Enables monitoring of donation and delivery status.
- Reduces manual record keeping.
- Improves transparency and efficiency in the donation-management process.

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13. Technologies Used

- Frontend: React.js, HTML, CSS, JavaScript
- Backend: Python, FastAPI
- Database: SQL-based database
- API: REST APIs
- Version Control: Git and GitHub
- Development Assistance: AI-assisted development tools

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14. References

1. "KindNest – An Essentials Connect," 2025 International Conference on Cognitive, Green and Ubiquitous Computing (IC-CGU), 2025. DOI: 10.1109/IC-CGU67042.2025.11338008.
2. "Real-Time Blood Donor Prediction and Campaign Optimization using DNN-Autoencoder and MERN Architecture," 2025 3rd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS), 2025. DOI: 10.1109/ICSSAS66150.2025.11080959.
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